

JOTUN PAINTS VIETNAM CO., LTD.

ELECTRICAL ENGINEERING SPECIFICATION FOR TELECOMMUNICATION SYSTEM

JOTUN VIETNAM-GREEN FIELD PROJECT

JOTUN JOB NO. 2017-2340

TVC JOB NO. VD-156

DOCUMENT NO. VNHG-EL-013

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1. GENERAL

1.1 Purpose

This specification covers the minimum requirements for documentation, design, manufacturing, construction, testing and preparation for shipment of the Telecommunication equipment of *Jotun VietNam-Green Field Project*, located in Hiep Phuoc Industrial Zone, Nha Be Commune, Ho Chi Minh City, Viet Nam.

1.2 Code and standards

- Codes and Standards shall include the following as :
 - TIA Telecommunications Industries Association.
 - EIA Electronic Industries Association.
 - EIA/ TIA 568B Commercial Building Telecommunication Cabling.
 - EIA/ TIA 568B Commercial Building Telecommunication Cabling Standard.
 - EIA/ RS 232C Interface between Data Circuit- Terminating Equipment Employing Serial Binary Data Interchange.
 - IEC 61158 Fieldbus Standard for Use in Industrial Control Systems.
 - IEC 62305 Protection Against Lightning.
 - IEEE 802.3af Standard for Power over Ethernet.
 - IEEE 802.3u Fast Ethernet (100BTX, 100BT4, 100BFX).
 - ITU International Telecommunication Union Standards.

1.3 Units and language

Unless otherwise specified metric unit (SI) system shall be used for all drawings and documents to be submitted by vendor.

All documents shall be written in the English language.

2. SERVICE AND ENVIRONMENTAL CONDITIONS

2.1 Site Conditions

For selection and provision of electric equipment, the following climatic conditions shall be taken as bases.

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- Ambient temperature
- Highest temperature : : 40°C
- Lowest temperature : : 20°C
- Relative humidity
- Maximum humidity : : 90%
- Altitude : : Less than 1000m above sea level.

2.2 Design Temperatures

Electrical equipment and materials to be installed above ground shall be design to give the specified guarantee rating at the maximum ambient design temperature of 40°C and humidity 90% unless otherwise agreed in writing.

2.3 Power Supply Conditions

Electrical power supply shall be of 230V AC, 50Hz, 1 phases, 3 wires. All equipment and materials to be required by this requisition shall be so designed as to operate satisfactorily with the following conditions and variations in the power supply.

- Rated voltage :230VAC Auxiliary Voltage, 1 Phase + N+ PE
- Voltage fluctuation :From -10% to +5% of Rated voltage
- Frequency :50Hz
- Frequency fluctuation :± 0.5Hz

3. SCOPE

- The telecommunicatoin system completed with all furnished components ready for installation. The telecommunicatoin system shall consist of the following equipment
 - Access switch **NOT COMPLIED**
 - Data outlet.
 - IP telephone (IP telephone shall be selected & supplied by Owner)
 - Patch panel.
 - 19'' rack.
- Accessories shall be included as required for proper operation.

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- Spare parts for construction and start-up shall be included in the scope of supply.
- Supervisor for commissioning and testing of Telecommunication System at site.
- Special tools for installation and commissioning (if any).
- All technical document and/or samples of material and equipment for approval.
- Shop inspection and test
- The Vendor of the equipment covered by this specification shall be responsible for coordination with other equipment which interfaces mechanically or electrically with his work.
- MEP contractor must be made coverage calculation document of wireless access point and submitted to Owner to finalize final location.

4. DRAWING AND DOCUMENT REQUIREMENT

Vendor shall submit to purchaser the drawings and documents as called for in this specification and in the purchaser's requisition. Purchaser's approval for such drawings and documents shall not relieve vendor of their responsibility to meet all requirements of the purchase order.

Vendor shall submit the following drawings and data in hard copy and electronic file (document in word or excel and drawing in AutoCAD), but not limited, as follows.

Drawing and document for approval / review

- Specifications and catalog
- Outline dimension drawings, including dimension of foundation, IP, color, thickness, loading data (weight), material specification etc.
- Bill of material and apparatus list.
- Manufacturing Schedule.
- Instruction Manual.
- Vendor print index and schedule.
- Internal/ External connection diagram.
- Schematic Diagram.
- Coverage calculation document of Wireless Access Point.

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5. PAINTING

Equipment shall be cleaned, primed and finish paint in accordance with applicable standards and manufacture's standard practice.

6. INSPECTION AND TESTS

The Telecommunication control Panel shall be tested according to applied standard and Manufacturer's standard routine.

7. PACKING AND SHIPPING

Packing and shipping shall be manufacturer's standards.

8. DESIGN BRIEF AND DATASHEET FOR TELECOMMUNICATION SYSTEM:

COMPLIED 8.1 Design brief and function.

Wireless access has been implemented to provide safe and trouble free LAN access for Jotun's employees and easy Internet access for external guests, customers and consultants. The WLAN may also provide access to IP telephony (VoIP).

The Telecommunication System shall have following capabilities:

- All access points must be dual radio 5Ghz with 1Gb LAN interface
- All access points must be a **minimum** of 802.11n compliant, preferably 802.11ac
- All access points must be connected to 1Gb switches

COMPLIED In addition

- Wireless clients must be connected to a dedicated VLAN
- Guest networks must be connected to a separate, segregated VLAN

8.2 Attachment documents.

Attachment – 01: Data sheet for communication system.

Attachment – 02: Wireless Network Implementation.

Attachment – 03: IT Infrastructure for new sites.

Attachment – 01

DATA SHEET FOR COMMUNICATION SYSTEM

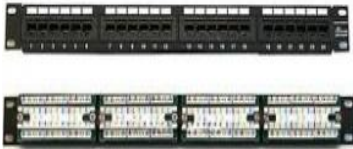
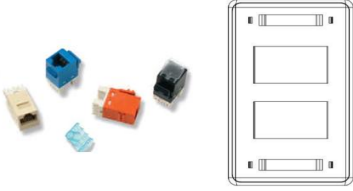
PROJECT NAME : JOTUN VIET NAM – GREEN FIELD PROJECT

DATA SHEET FOR TELECOMMUNICATION SYSTEM

ITEM NO.	DESCRIPTION	OUTLINE DRAWING (FOR REFERENCE ONLY)	APPLICATION	Remark
1	Switch 48-Port			
	- Suitable for power supply 230VAC, 50Hz			
	- General colour as per manufacturer standard			
	- Supported protocol: IPv6 and IPv4			
	- 48x10/100 Mbps Ports + 2 dual-purpose ports (10/100/1000 or SFP), LAN Base			
	- Operating temperature: 32° to 104°F (0° to 40°C)			
	- Operating humidity: 10% to 90%, relative, non-condensing			
	- Suitable for installation to 19" Rack			
	- Completed with Switch Rack Mount Bracket, power cable, data cable and other other accessories as required for a completely set.			
2	Switch POE 48-Port	COMPLIED WITH HPE SWITCH 		
	- Suitable for power supply 230VAC, 50Hz			
	- General colour as per manufacturer standard			
	- PoE Switch, Layer 2, Non- Stack Module			
	- Supported protocol: IPv6 and IPv4			
	- 48x10/100 Mbps Ports + 2 dual-purpose ports (10/100/1000 or SFP), LAN Base			
	- Operating temperature: 32° to 104°F (0° to 40°C)			
	- Operating humidity: 10% to 90%, relative, non-condensing			
	- Suitable for installation to 19" Rack			
	- Completed with Switch Rack Mount Bracket, power cable, data cable and other other accessories as required for a completely set.			
3	Switch 24-Port			
	- Suitable for power supply 230VAC, 50Hz			
	- General colour as per manufacturer standard			
	- Supported protocol: IPv6 and IPv4			
	- 24x10/100 Mbps Ports + 2 dual-purpose ports (10/100/1000 or SFP), LAN Base			
	- Operating temperature: 32° to 104°F (0° to 40°C)			
	- Operating humidity: 10% to 90%, relative, non-condensing			
	- Suitable for installation to 19" Rack			
	- Completed with Switch Rack Mount Bracket, power cable, data cable and other other accessories as required for a completely set.			
4	Switch POE 24-Port	COMPLIED WITH HPE SWITCH 		
	- Suitable for power supply 230VAC, 50Hz			
	- General colour as per manufacturer standard			
	- PoE Switch, Layer 2, Non- Stack Module			
	- Supported protocol: IPv6 and IPv4			
	- 24x10/100 Mbps Ports POE+ 2 dual-purpose ports (10/100/1000 or SFP), LAN Base			
	- Operating temperature: 32° to 104°F (0° to 40°C)			
	- Operating humidity: 10% to 90%, relative, non-condensing			
	- Suitable for installation to 19" Rack			
	- Completed with Switch Rack Mount Bracket, power cable, data cable and other other accessories as required for a completely set.			
5	Rack 42U (Free Standing) _ Series Server Cabinet			
	- Rack 19" 42U, 800 x 2000 x 1000 (mm) (WxHxD)			
	- Material: Steel sheet. Frame :2mm thickness. Front door 1,5mm: thickness. Side door: 1,2mm thickness. Rustproof, Electro Static painting.			
	- Front door: 80% Flat Perforate steel 1,5mm with lock for security			
	- Back door: Flat steel 1,5mm with two small half door			
	- Incoming/outgoing cable shall be located at rear bottom and top of Rack.			
	- Static loading: 1000kg			
	- Colour: Black			
	- Standard: ANSI/EIA 310-D, Type A, 1U = 44.45 mm (1.75") DIN 41494 ,BS5954 Part 2,IEC 60297-1, IEC 60297-2			
	- Completed with ventilation fans, horizontal cable management, power distribution unit and other accessories as required for a completely set.			
	- Completed with fix Shelf Depth 850, Black			
	- Name plate for LAN Rack shall be stainless steel			

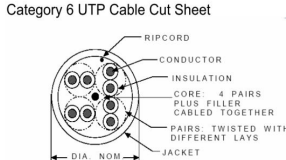
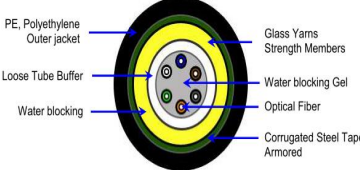
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DATA SHEET FOR TELECOMMUNICATION SYSTEM

ITEM NO.	DESCRIPTION	OUTLINE DRAWING (FOR REFERENCE ONLY)	APPLICATION	Remark
6	Rack 27U (Free Standing)			
	- Rack 19" 27U, 600 x 1420 x 1000 (mm) (WxHxD)			
	- Material: Steel sheet. Frame: 2mm thickness. Door: 1,2mm thickness. Rustproof, Electro Static painting.			
	- Front door: Acrylic clear visible inside with lock for security.			
	- Incoming/outgoing cable shall be located at rear bottom and top of Rack.			
	- Static loading: 100kg			
	- Colour: Black			
	- Standard: 1U = 44.45 mm (1.75")			
	- Completed with ventilation fans, horizontal cable management, power distribution unit rack mounted and other accessories as required for a completely set.			
7	Rack 15U (Wall Mounted Enclosure)			
	- Rack 19" 15U, 600 x 780 x 450 (mm) (WxHxD)			
	- Material: Steel sheet. Frame: 2mm thickness. Door: 1,2mm thickness. Rustproof, Electro Static painting.			
	- Front door: Acrylic clear visible inside with lock for security.			
	- Incoming/outgoing cable shall be located at rear bottom and top of Rack.			
	- Static loading: 100kg			
	- Colour: Black			
	- Standard: 1U = 44.45 mm (1.75")			
	- Completed with ventilation fans, horizontal cable management, power distribution unit rack mounted and other accessories as required for a completely set.			
8	Fiber Optic Rack Mount Patch Enclosure			
	- Enclosure pre-loaded with SC Adapters			
	- Completed with cable strain relief and management rings			
	- Suitable for installation to 19" Rack			
	- Suitable for multimode fiber optic cable			
	- Fiber 12 core fiber, 6 duplex port used			
	- General colour as per manufacturer standard			
	- Completed with cable gland, splice tray and other accessories as required for a completely set.			
9	Cat 6 Patch Panel			
	- 24 Port 1U			
	- Completed with 24 modular jack and other accessories as required for a completely set.			
10	Cat 6 Patch Panel			
	- 48 Port 2U			
	- Completed with 48 modular jack and other accessories as required for a completely set.			
11	Data outlet (Double Face)			
	- Double Gang RJ45 Outlet			
	- Gang RJ45 Outlet			
	- Completed with 02 Cat 6 Modular Jack RJ45			
	- Face plate of Cat 6 Modular Jack (with label & shutter)			
	- Completed with plastic faceplate label and other accessories as required for a completely set.			
12	Data outlet (Single Face)			
	- One Gang RJ45 Outlet			
	- Gang RJ45 Outlet			
	- Completed with 01 Cat 6 Modular Jack RJ45			
	- Face plate of Cat 6 Modular Jack (with label & shutter)			
	- Completed with plastic faceplate label and other accessories as required for a completely set.			

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DATA SHEET FOR TELECOMMUNICATION SYSTEM

ITEM NO.	DESCRIPTION	OUTLINE DRAWING (FOR REFERENCE ONLY)	APPLICATION	Remark
13	Wireless Access Point Meraki MR74		Warehouse Area	
	+Complied with :			
	-Meraki Dual-band Omni Antennas (4 poles antennas)			
	-Meraki MR74 Cloud Managed AP			
	-Meraki MR Enterprise License, 5YR			
14	Wireless Access Point Meraki MR33		Office Area	
	+Complied with :			
	-Meraki MR33 Cloud Managed AP			
	-Meraki Dual-band Omni Antennas			
	-Meraki MR Enterprise License, 5YR			
15	Cat 6 Modular Plug RJ45			
	- Modular Plug Cat 6, 8 Pins, 23-24AWG, for Solid Cable, 6-7mm			
	- Completed with Modular Plug Boot RJ-45, Blue			
	- General colour as per manufacturer standard			
16	Cat 6 Patch Cord			
	- 4 Pair stranded UTP cable			
	- CM Jacket			
	- UTP Cat 6 SL Series Patch Cord 24AWG 10FT (3M)			
	- Colour: Blue			
17	Fiber Optic Cable Patch Cord			
	- Duplex SC to Duplex SC, OM3, 10FT (3M).			
	- Colour: Aqua			
18	Category 6 UTP Cable			
	- Cat 6 unsheild twisted pairs cable			
	- Conductor: 23 AWG Solid Copper			
	- Insulation: Polyethylene (PE)			
	- Jacket: LSHF			
	- Fire Rating: CM			
	- Standard: TIA/EIA-568-B.2-1 Category 6			
	- Outer jacket colour: Blue			
19	Outdoor Fiber Optic Cable			
	- Multi-mode Fiber Optic (6C) Outdoor, OM-3, 50/125 micro metre			
	- Completed with corrugated steel tape armored and loose tube buffer			
	- Outer jacket colour: Black			

Attachment – 02

WIRELESS NETWORK IMLEMENTATION

WIRELESS NETWORK IMPLEMENTATION

Document info

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Purpose

The purpose of this document is to outline the standards and methods employed when implementing a wireless network in a Jotun site.

Introduction and overview

Wireless LAN (WLAN) access is an essential feature which provides local mobility and access when available to Jotun's users. But unprotected wireless access can be open to improper use or severe hostile attacks. It is important that WLAN is implemented in a controlled and secure way.

The purpose of this document is to describe the implementation of a wireless network and the standards to be followed in any Jotun site.

Wireless access has been implemented to provide safe and trouble free LAN access for Jotun's employees and easy Internet access for external guests, customers and consultants. The WLAN may also provide access to IP telephony (VoIP).

At the time of writing this policy we currently have managed wireless networks utilizing 802.1x standard (Radius authentication/AD integration) in 72 locations in 40 countries with a total of approximately 650 access points.

These managed networks are mostly comprised of 3 different systems; HP 700 series controllers along with the MSM family of access points, a smaller number of Cisco 2500 series controllers with the LAP/CAP family of access points and the Cisco Meraki MR suite of Access Points.

COMPLIED Wireless standard – General

Having had the opportunity to test several different systems including those currently in place we have chosen the Cisco Meraki suite of products as the new Global Standard.

The benefits of the Meraki products are many and varied; the main benefit being that the product is "cloud based" meaning that there are no physical controllers employed in the deployment and management process. All management tasks are performed via a single web portal which greatly simplifies the process and removes the need to have knowledge of many different systems, firmware versions, local configurations etc.

In addition, with Meraki we are now able to offer a tiered management structure, whereby Global policies, licenses, administrators etc. are centrally created and administered whilst local administration is set at a local and/or regional level allowing local management without compromising Global security.

The full range of current Meraki products can be found [here](#)

COMPLIED New challenges

With the increase of personal devices in use, (smartphones, tablets etc.) and with the advent of O365 there is now a greater requirement than ever for better performance and coverage to improve the end-user experience.

Whilst some of the current wireless equipment is still adequate, there will be almost certainly be a need to supplement and upgrade existing equipment. If the legacy equipment is not at the minimum standard required, it must be replaced as soon as possible.

Any new sites should invest in the best equipment possible to ensure the best end-user experience as well as to “future-proof”.

COMPLIED General equipment standards

- All access points must be dual radio 5Ghz with 1Gb LAN interface
- All access points must be a **minimum** of 802.11n compliant, preferably 802.11ac
- All access points must be connected to 1Gb switches

In addition

- Wireless clients must be connected to a dedicated VLAN
- Guest networks must be connected to a separate, segregated VLAN

Support

All new wireless installations must follow this standard on configuration and hardware to get support from Group IT. Existing installation will get best effort support from Global IT as the systems differ not just between vendors but also between firmware release versions.

We strongly recommend that non-Meraki based systems – such as HP, Cisco etc. – take regular backups of their controller configurations to enable a quick restore in the event of failure or configuration error.

COMPLIED Internal

For confidentiality (data encryption) 802.11i (or WPA2) is used.

For authentication (log in) we are using 802.1x using Extensible Authentication Protocol (EAP) through AD.

Jotun’s own employees connect to the WLAN with no manual intervention. The Client’s AD credentials are authenticated and if authorized, an encrypted tunnel is set up between the user and the LAN on the WLAN.

COMPLIED Jotun Guests

A guest user is **only** allowed access to the Internet, no Jotun internal resources. Guest users can connect in several different ways, including requesting a time-limited user-id and password, preprinted voucher etc. We are currently working in conjunction with the IT Security project team to determine the best way to handle this with a view to implementing Globally.

We hope to create self-service portals whereby guests create their own user account which is authorized by a member of Group IT - this is currently under discussion.

Any guest networks currently using static passwords must be changed monthly.

COMPLIED Jotun devices and personal devices

We have found that many users are accessing network resources via the InternJotun2 SSID. Because this wireless network is designed specifically to accommodate business equipment there is the potential to seriously impact available bandwidth. With that in mind we are currently working towards providing a separate Global wireless network that will allow access to in-house resources whilst minimizing the impact on local bandwidth by using either a low-cost Internet link or the site “main” link with a bandwidth restriction.

We are also exploring the possibilities of scanning non-Corporate resources for security software and barring any devices that don’t meet these basic criteria.

COMPLIED Active Directory

The security in the wireless network is based on pre-defined access privileges given to users via Active Directory (AD). The Domain Admin decides who to give WLAN access to, put these users and machine accounts in a manageable structure, e.g. dedicated WLAN group, and then apply a specially designed group policy (GPO) to this group. The GPO is then applied on the users machines on their first domain logon, or manually by doing a **gpupdate** at the command prompt (Note: This process has to be done when wired on the internal network).

Next time the user starts up their machine with wireless activated, the applied GPO will force a connection to the pre-defined SSID (InternJotun2), by authenticating the machine via RADIUS/AD. The machine will after a successful authentication, receive an IP address via DHCP.

All this happens before the windows login prompt appears, so the next step is user authentication via the normal domain logon procedure.

Network Name (SSID)	Network Broadcasts its SSID
Internjotun (or 1-2)	True
Security Settings	
Authentication	WPA2
Encryption	AES
Use 802.1X	Enabled
Pairwise Master Key (PMK) Caching	Enabled
PMK Time-to-Live (minutes)	720
Number of Entries in PMK Cache	128
Use Network Pre-authentication	Disabled
IEEE 802.1X Settings	
Computer Authentication	User re-authentication
Maximum EAPOL-Start Messages Sent	3
Held Period (seconds)	60
Start Period (seconds)	60
Authentication Period (seconds)	30

Figure 1 - Screenshot from the AD global WAN policy

COMPLIED Radius

The Radius server is configured by Global IT Server Services team through a request to GSD. We use the AD domain controllers for Radius (802.1x) authentication.

By utilising F5 Big IP technology we have successfully centralised the Radius process authenticating Radius requests via a pool of virtual servers which in turn are load balanced across a pool of Radius Servers, resulting in both high availability and increased performance.

New installations

Having decided to purchase equipment the following procedures should be followed:

- NA • Make a site survey (see 1.2.1 for details of this) to determine the equipment that is appropriate for the environment where it will be installed. Having made the survey and determined the equipment, place your order with the local Meraki vendor, if there is not a local vendor please contact Network Team who can if necessary purchase on your behalf.
- COMPLIED • When purchasing hardware, you must also purchase a Dashboard Support licence, although several terms are available we recommend you choose a 5-year licence; this works out cheaper in the long term and will be sufficient for the equipment lifecycle.
- COMPLIED • When the order is processed, you will receive a Dashboard Support licence key, it is imperative that you pass this key to Network Team to add to the Global inventory, under no circumstances should you attempt to create your own organisation.
- COMPLIED • At the same time, you should also receive a Meraki Order number, this must also be passed to Network Team to allow us to add the new access points to the Global inventory. If the order number is not received, you must pass the serial numbers to Network Team instead.
- COMPLIED • When the licence key and serial numbers/order number are received, we will add the licences and devices into the correct repository's, create your networks and add the access points to the new networks.
- COMPLIED • At this stage, you should advise Network Team of your local Administrators to enable us to grant access to the new network.

You will now have a fully functioning wireless network with the correct SSIDs ready to go, all you need do is to physically install and cable the access points and ensure the correct VLANs are defined on your switches. Once the access points are installed they will immediately work.

***** Please note it is extremely important that Meraki equipment is NOT mixed with other equipment at the same site as this will seriously affect roaming.
The only exception to this rule is where different equipment is in separate buildings where the distance between buildings is sufficient to cause the connection any connections to drop. *****

NA Wireless planning

Make a proper survey of the site, if possible obtain blueprints/CAD drawings to aid in the design process and if necessary employ a specialist company to conduct the survey.
Consider the following:

- Coverage areas – indoor/outdoor have different requirements and require different solutions
- Building layout and construction materials
- Client numbers and mix (smartphones, tablets, laptops etc.)
- Client density distribution
- Critical applications and business processes
- Wired network integration
- Potential interference from external sources such as other radio devices etc.
- WLAN channel overlapping¹

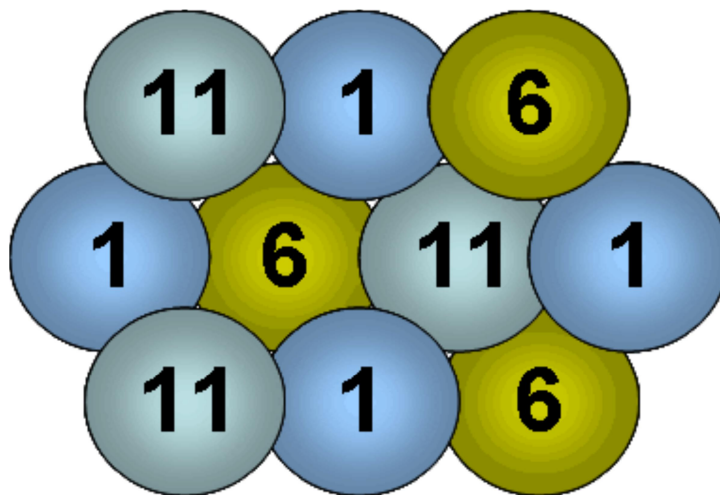


Figure 2 - Typical spread of channels in a multi access point environment

NA Site maps

The site maps on the following pages illustrate the different zones within a standard production site and highlight the need for different hardware in different locations such as Office, Production areas, Finished Goods Warehouses etc.

¹ The Meraki solution has built-in channel handling. This is to ensure that we don't have interference from neighbouring access points, but at the same time to ensure roaming.
Under normal circumstances no configuration of the radio channels is required, however special circumstances such as external interference from other radio devices may require some manual intervention, in these instances Network Team will be able to advise and assist.

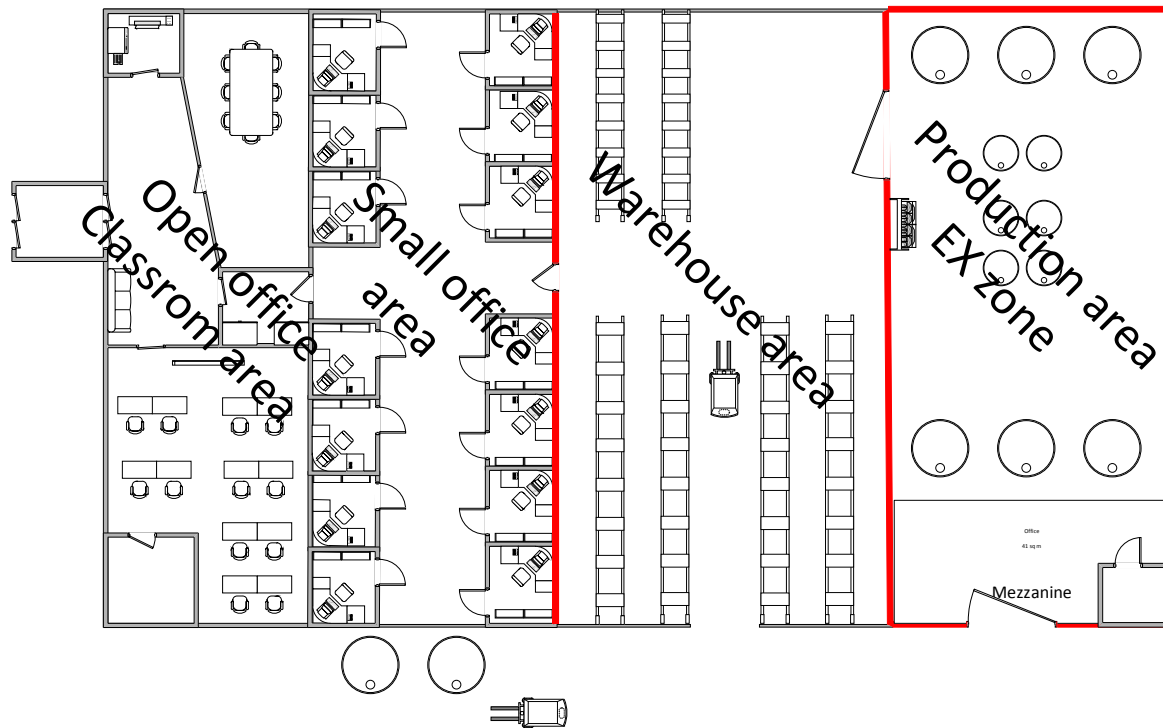


Figure 3 - Site map

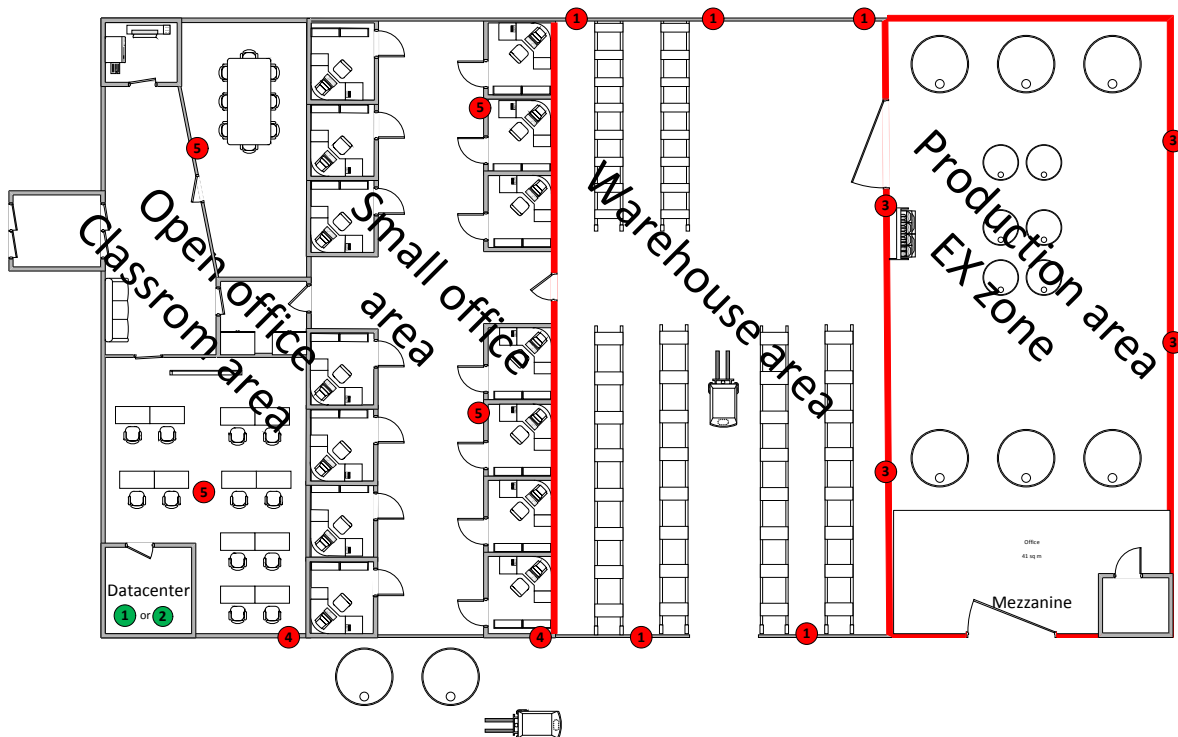


Figure 4 - Site map with access points

Associate the numbers of the access points with the different models on the Meraki website.

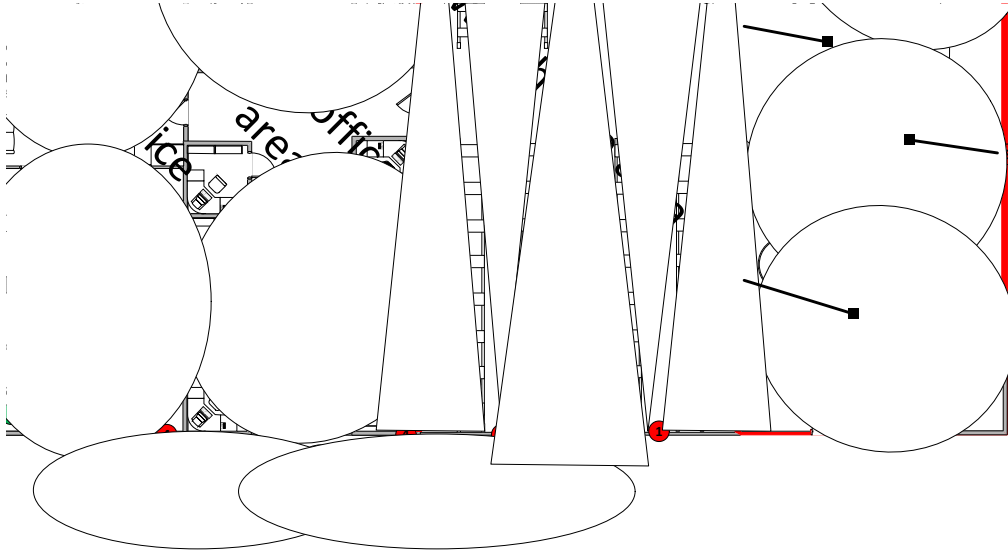


Figure 5 - Site map with access points and WIFI areas

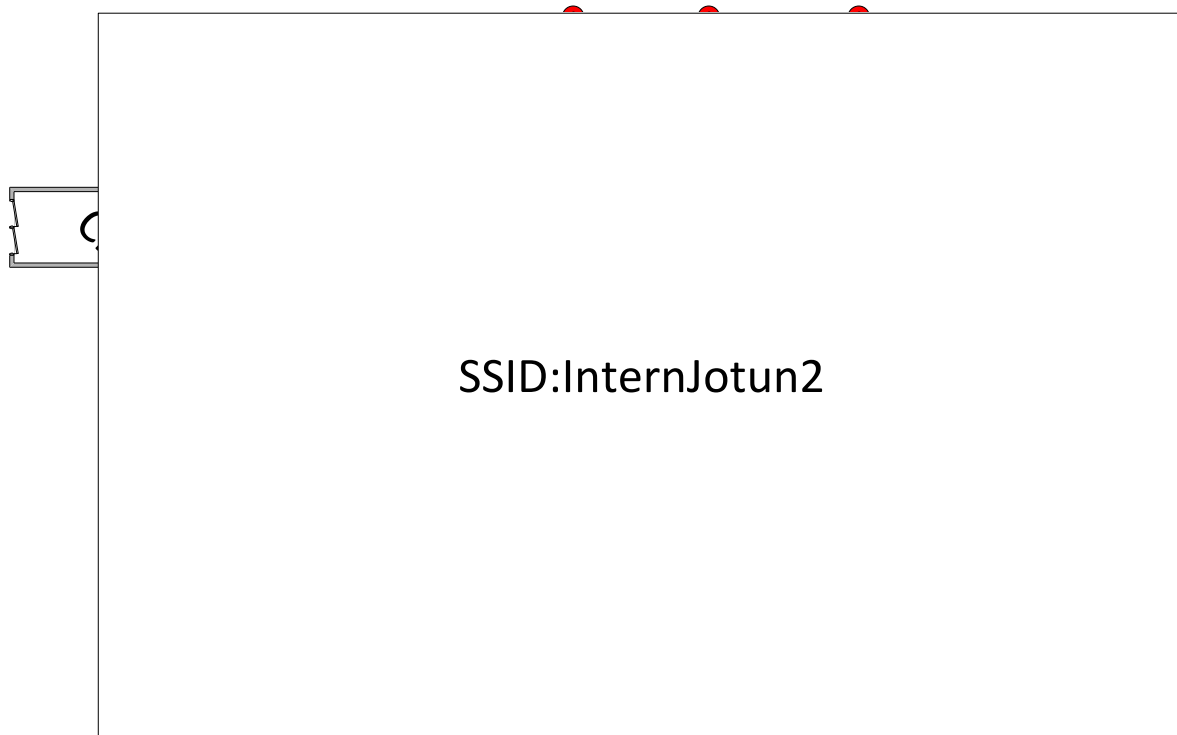


Figure 6 - Site map with SSID InternJotun2

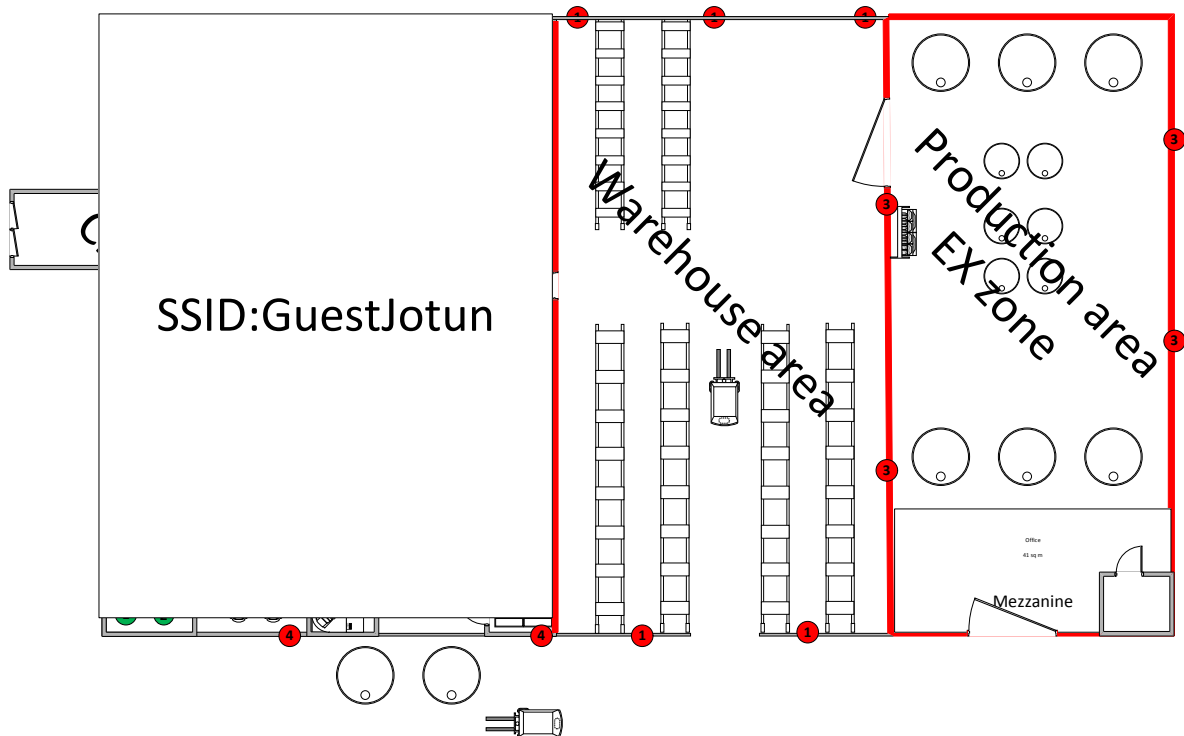


Figure 7 - Site map with SSID GuestJotun

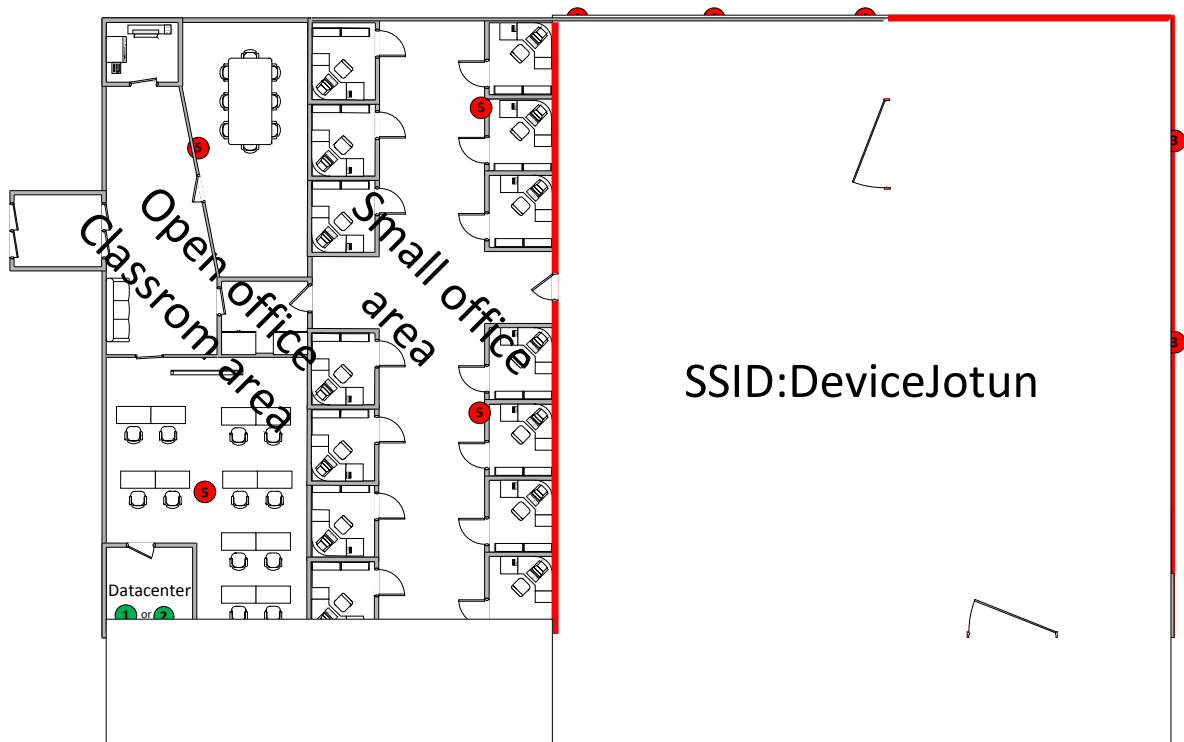


Figure 8 - Site map with SSID DeviceJotun

COMPLIED Maximizing your installation – best practices

The procedures outlined below apply not just to new installations but to existing equipment (Meraki or otherwise) and should be followed to ensure that the best levels of performance are obtained.

- Ensure wireless traffic is on a dedicated VLAN.
- Enable “Band-Steering” to ensure maximum usage of 5Ghz radio band – 5Ghz radios can provide up to 8x the capacity of 2.4Ghz
- Place access points where users are located – in room placement is always best, try to avoid hallways where possible.
- Disable low data rates such as 802.11b - this can significantly impact network performance
- Identify and if possible remove any possible sources of external interference. If removal is not possible utilise different channels to minimise the impact.
- Minimise the number of SSIDs – fewer SSIDs means less network overhead
- Set and enforce maximum devices per radio where applicable
- Remove SSIDs still using TKIP encryption – less secure and lower data rate.
- Analyse traffic to determine what is bandwidth hungry and shape as necessary.

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Attachment – 03

IT INFRASTRUCTURE FOR NEW SITES

IT INFRASTRUCTURE FOR NEW SITES

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IMPORTANT!

The contents of the document could be changed without notifying the reader. The only valid edition of the document is the one available in the IT Library.

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Purpose

The purpose of the document is to outline the standards, products and general considerations to be employed when installing IT Infrastructure in a new Jotun site.

Introduction & Overview

IT Infrastructure is an essential requirement for any new or existing site; without the appropriate infrastructure it is impossible to deliver any kind of IT service. Whilst existing sites often evolve their infrastructure “organically” over a period of time to suit their specific needs, the building of new sites offers the opportunity to both fulfil the current business requirements and to build for the future.

With the Jotun philosophy of organic growth, we see new sites opening regularly, and each site brings its own set of challenges. For the purposes of this document we will assume that the new site is a factory with multiple buildings all requiring access to Jotun resources and the Internet, however many of the principle and standards discussed can easily be applied to a small regional sales office containing a handful of staff.

General considerations

Project planning

A successful implementation depends on proper planning at the outset; this document intends to provide the relevant points on which to build a detailed plan. Whilst not specifically mentioned in this document (which concentrates more on the technical/physical aspects of an implementation) it's important to ask these kinds of questions at the outset:

Budget – is it realistic and can we stay within it?

Timescales – are they realistic and attainable?

Purchasing – are the right products available and do the lead-times fit the timetable?

Order of events – what is dependent on what and do we have everything in the correct order?

Resources – do we have the necessary knowledge in house or do we need to “buy in”?

Documentation – what type of documentation and who do we share with?

Communication – who are the relevant parties and is everyone “in the loop”?

Responsibilities – does everyone know their roles?

Testing – has a rigorous and detailed test plan been made?

There are plenty of helpful corporate tools available for project planning; access to these tools can be found in Join under “Resources>Forms & Templates”

Overall Topology

Overall topology is the first consideration and for our purposes describes the physical network linking the various buildings. The usual choices are Ring, Star and Hybrid-Star with each having their own advantages and disadvantages. In the majority of cases a Star network is appropriate with the building locating the server room being the centre of the star; however the final choice is dependent on a number of different elements including:

- Number of buildings requiring infrastructure
- Distances between buildings
- Network resilience
- Future proofing
- Cost/budget

The single most important consideration at this stage of the project is the distance between buildings. If the distance is less than 100m then CAT6 cabling can be considered, however any distance greater than 100m will require fiber cabling due to the limited range of CAT6.

Consider suggesting that ALL building-to-building uses fiber.

Network Resilience

All networks should have resilience incorporated into the design to minimise disruption to IT services. In the case of a Ring topology, network protocols such as RPR (Resilient Packet Ring) can be implemented. In the case of a Star network, multiple physical links in conjunction with STP (Spanning Tree Protocol) or RSTP (Rapid Spanning Tree Protocol) to prevent network loops should be used.

Types of Fiber

There are two main types of fiber cabling – Single mode & Multimode; their advantages and disadvantages are outlined below:

Type	Advantages	Disadvantages
Single mode	Good for Ring topology	More expensive than Multimode fiber
	Range up to 40Kms (@10Gb/s)	Transceivers more expensive than Multimode
Multimode	Good for Star topology	Unsuitable for Ring topology
	Cheaper than Single mode	Less range than Single mode fiber
	Range up to 550m (@10Gb/s)	

In the majority of cases we would recommend the use of OM3 or OM4 fiber; however if the range between buildings exceeds 550m or bandwidth greater than 10Gb/s is required then Single mode fiber should be considered.

Types of Ethernet

There are many types of Ethernet cabling available, however at this time we strongly recommend the use of CAT6 cabling in all instances.

Other considerations

Other more general items need to be given careful consideration and differ not just from site to site, but often from building to building; these include:-

- Number of LAN points - data & telephony, are there sufficient?
- Location of LAN points – consider the placement of Wireless Access Points and the installation of LAN into specialist areas such as Video Conference suites and Copier Rooms etc.
- Number and location of power sockets – the same considerations should be given to power as to LAN points.
- Location of network cabinets in remote buildings – they must be at a workable height whilst safe from accidental damage and should be lockable. Also consider the provision of power to these cabinets; depending on local conditions, A/C may also be required.
Also consider the size of the cabinets – will they accommodate all the switching required and if required a UPS?
- Manholes & conduits will be needed for the installation of fiber and CAT6 cables, it's imperative that these are installed in the correct locations. Also remember to ensure “draw cords” are installed in these conduits to enable easy installation of cabling.

NA Server Room

The Server Room is the single-most critical area where infrastructure is concerned and as such requires special focus and attention.

The Server Room should be kept locked at all times and access restricted to authorised staff only.

Cabling

Because the server room is generally the “nexus” of the cabling infrastructure – particularly fiber – the proper conduits etc. mentioned in 1.1.5 must be present.

Proper network racking should be installed along with the appropriate fiber termination and CAT6 termination panels.

Air Conditioning

Proper air conditioning is essential in any server room to ensure that equipment is kept within proper operating temperatures. Specifications vary depending on factors such as room size, equipment in use, equipment power consumption etc. Various tools are available online to assist with the sizing of air conditioning units; vendors will also be able to assist. We consider it best practice to install a backup air conditioning unit that is capable of maintaining the correct environment to protect against failure of the primary unit. Optimum temperature is agreed at around 22 degrees.

Power

Power must be supplied to the server room via a UPS to provide continuous protection against power failures/brownouts etc. The type of UPS installed depends on a number of factors including:-

- Criticality of the systems under protection – i.e. mission critical servers, surveillance systems, network firewalls etc.
- Number of servers/appliances and their total power consumption
- Battery runtime required
- Form factor – tower, mini-tower, rack etc.

There should be close liaison between IT & the team responsible for electrical work to ensure the correct UPS is chosen; any errors at the design stage can be costly and difficult to remedy once the installation is complete.

To ensure maximum protection we would recommend a minimum of 2 x PDUs (power distribution units) are used as this will allow connection of dual-PSU machines (such as servers, SANs etc.) to multiple PDUs thereby reducing the risk of downtime in the case of a PDU failure.

For the same reason we would also recommend that each switch in a stack is connected to a separate PDU.

In the instance of a major factory installation it would be best practice to install 2 completely separate UPS's.

Fire suppression systems

Fire suppression systems are essential to the server room itself; a fire suppression system will automatically extinguish a fire without the need of human intervention.

The two main types are inert gas systems that generally use Argon/Nitrogen to reduce the O2 level in the server room and chemical/synthetic systems that cool the room to reduce the temperature. There are pros and cons to both systems and we suggest that expert advice from a specialist vendor is obtained to determine the most suitable system for the site.

Please note that the design standards for fire suppression are carried out with strict guidelines as the fire suppression agents used can be extremely dangerous if not designed correctly.

Flooring

Raised/false floors are important for a number of reasons including enhanced cooling, enhanced cable management and reduction of trip hazards. If water piping is required in the room for fire suppression, A/C systems etc. it is far safer to run these pipes under the floor than inside the ceiling. For small/medium sized installations such as Sales Offices a raised floor is not strictly necessary.

Other considerations

The small details can make a big impact once implementation is complete and thinking the organisational details at this stage can make a big later in the project, these include:-

- Colour schemes – There is no standard currently in place but implementing your own colour scheme can reduce the time spent in the main patch cabinet. For instance Red = Data, Black = Telephony, Yellow = Wi-Fi etc.
- Labelling – Label as many items as possible to help identify them later. Items such as critical links, fiber terminations, stacking cables etc.

WAN Connectivity

WAN components

NA Internet links

For a major site such as a new factory it is essential that there is resilience in the WAN and to this end we strongly advise that a minimum of two Internet links (preferably fiber where possible) are provisioned. For further resilience we recommend that these links are provided by different ISPs. It is important that Network Team are involved in the design process at this stage to advise on specific designs on a site by site basis.

The Internet routers should be powered by separate PDUs inside the server room and configured for connectivity to multiple firewalls (clustering).

As well as providing High Availability resilience, network traffic can be split into VPN (Matrix, Collaboration traffic etc.) and non-VPN (Internet browsing etc.) and routed across the individual links maximising and improving traffic throughput and performance.

In addition it's also useful to think about Guest traffic at this stage; for instance can we route traffic generated by non-Jotun people and personal devices directly out of the network via a separate cheap DSL link thereby reducing the overhead on the network generally?

NA Firewalls

Network Team will provide the recommendations and in the majority of cases (depending on local purchasing factors) the physical firewalls for installation. For a new factory/major site we would always recommend that the firewalls are configured as a two node cluster thereby providing the maximum amount of network resilience.

As with the Internet routers the firewalls should be powered by separate PDUs inside the server room and cabled into both sides of the core switch. (*Core switch configurations will be discussed later.*)

LAN components

COMPLIED Switches

As a general rule Global IT recommend the use of HP/Aruba switches; there is a large range of switches available to suit any sized site in almost any kind of configuration. The size and type of the site and the network topology discussed in section 1.1.1 will determine the switches and ancillaries required.

If there is no favourable HP/Aruba vendor or maintenance agreement available in a particular region/country then the equivalent Cisco or Meraki switches would also be approved for use. Meraki switches have the added benefit of being managed from the same “dashboard” as the WiFi equipment that is the current Jotun standard.

NA Core switches

A core switch is a high-capacity switch generally positioned within the backbone or physical core of a network. Core switches serve as the gateway to a wide area network (WAN) and generally provide the final aggregation point for the network.

Server infrastructure and wireless controllers would typically connect directly to the core switch although depending on the size of the new infrastructure, “aggregation switches” may also be used; these switches house the infrastructure and in turn connect directly to the core switch.

Because the core switch is a backbone device and central to the operation of the network it is essential that resilience is built into the design.

Please see some basic examples of core switch models and resilience scenarios below:-

Size of site	Core Switch Type	Resilience employed
Small factory <100 users	HP/Aruba 2920	2 x switches in stack configuration with resilient cabling to clustered firewalls
	HP/Aruba 5406	Single switch with redundant PSU and Management modules plus <4hrs maintenance agreement
Medium factory <250 users	HP/Aruba 5406	Single switch with redundant PSU and Management modules plus <4hrs maintenance agreement
	HP/Aruba 5406	2 x switches paired with resilient cabling to clustered firewalls
Large factory >250 users	HP/Aruba 5406/5412	2 x switches with redundant PSU and Management modules plus <4hrs maintenance agreement paired with resilient cabling to clustered firewalls. Stacked aggregation switches for server infrastructure.

The table above is for example purposes only; each installation is slightly different and will have different requirements that should be handled on an individual basis.

We currently recommend the 54xx and 29xx series switches for use as core switches. The 29xx is ideal for smaller sites when configured as a stack, whilst the modular nature of the 54xx provides the flexibility and scalability required for larger installations.

Further information about the 54xx series of switches can be found here:-

<http://www8.hp.com/us/en/products/networking-switches/product-detail.html?oid=7074783#!tab=features>

COMPLIED Edge switches

Edge switches, are used to connect client devices, such as laptops, desktops, security cameras etc. to the network. For this reason, edge switches generally are considered less crucial than core switches to a network’s smooth operation.

Despite the fact that edge switches are not backbone devices, resilience is still important as they are the end users direct link to network resources and therefore have a direct effect on the end-user experience.

Some of the points that should be considered when choosing edge switches are highlighted below:-

- Number of users – how many switch ports are required?
- PoE – IP phones, Wireless Access Points need to be powered – how many PoE ports will be required?
- Core switch connectivity – What modules are needed to connect and can the switch accommodate them?
- Can the switches be stacked if necessary?

For general use we would recommend the HP/Aruba 29xx family of switches for use as edge switches. They support 10Gb uplinks, are available in a number of configurations and are a member of the same product family and run the same operating system as the 54xx range of core switches. Further information about the 2920 series of switches can be found here:-

<http://www8.hp.com/h20195/v2/getpdf.aspx/c04111401.pdf?ver=14>

Ancillary equipment

For the purpose of this document ancillaries are defined as additional add-on devices or modules that can be incorporated into core and edge switches to enhance/improve functionality. These include equipment such as GBIC transceivers, stacking modules, management modules etc.

Please see a list of some of the more commonly used ancillaries below:-

Part number	Description	Usage
J9538A	HP 8-port 10GbE SFP+ Module	54xx series fiber module for connecting edge switches @ 10Gb (requires transceivers)
J9829A	HP 5400R 1100W PoE+ Power Supply	54xx series power supply to provide redundancy/resilience
J9852A	HP X450 4-Post Rack Mount Kit	Rack mounting kit for 54xx series switches
J9731A	HP 2920 2-port 10GbE SFP+ Module	29xx series fiber module for connecting core switches @10Gb (requires transceivers)
J9150A	HP X132 10G SFP+ LC SR Transceiver	Transceiver to enable connection of core<>edge switches @ 10Gb
J9733A	HP 2920 2-port Stacking Module	Module to enable stacking of up to 4 x 2920 switches
J9534A	HP 24-port 1Gb PoE+ Module	Provides 24 ports of 1Gb PoE switching for 54xx series switches
J9827A	HP Management Module	Provides redundancy/resilience to 54xx series switches
J9283B	HP X242 10G SFP+ 3m DAC Cable	3mtr copper cable to connect core<>edge switches in same server room
J9734A	HP 2920 0.5m Stacking Cable	0.5mtr cable to enable switch stacking

COMPLIED VLANs

For the purpose of this document a VLAN is defined as a network of computers that behave as if they are connected to the same wire even though they may actually be physically located on different segments of a LAN. VLANs are configured through software rather than hardware, which makes them extremely flexible.

Advantages gained by implementing VLANs include the reduction of broadcast traffic thereby reducing network congestion and enhanced network security. Further benefits include the ability to

physically move equipment to different locations without any reconfiguration – this is particularly useful when moving from an old site to a new one.

Please see below an example of a recently implemented site and the various VLANs used along with their different roles.

Notice that in this example a /21 network has been allocated and sub-divided into a series of consecutive /24 subnets with two further IP subnets available for use as/if required.

We recommend this as best practice.

VLAN ID	IP Subnet	Name	Description/Role
100	10.111.232.0/24	INFRASTRUCTURE_VLAN	Used specifically for switch & wireless infrastructure (Default)
101	10.111.233.0/24	CLIENT_VLAN	Used specifically by end-users
102	10.111.234.0/24	VOIP_VLAN	Used specifically for IP Telephony
103	10.111.235.0/24	WIRELESS_VLAN	Used for internal wireless clients
104	10.111.236.0/24	GUEST_VLAN	Used for guest wireless clients
105	10.111.237.0/24	SECURITY_VLAN	Used specifically for internal security system – cameras, locks etc

Network team will advise on and recommend/assist in the use of VLANs as required as well as allocating the various IP subnets.

Wireless Networking

Wireless networking is covered in detail by document “0302 Wireless Local Area Network”
Please contact Network Team if you do not have this document.

Testing

General

The test phase of the project is essential for effective project completion, if testing is not thorough and rigorous, series issues could arise which would have been discovered during the testing phase.

As well as general functionality tests, it's important for the testing phase to be as realistic as possible so that real-world events are replicated, thereby highlighting any potential issues that can be avoided and allowing contingency plans to be made.

Because different countries have different lead-times on delivery of equipment, we strongly recommend that not only is all equipment ordered as early as possible, but on arrival is immediately inspected and tested.

It's always possible that either faulty or incorrect equipment has been delivered, therefore the earlier these potential problems are found the quicker they can be resolved without impacting on the overall timetable.

For example imagine that a batch of SFP fiber modules were delivered which were discovered to be incompatible with the chosen switches and the lead-time for replacement was 2-3 weeks.

Initial testing on arrival would ensure that the correct modules arrive without seriously impacting the overall project; if however initial inspection/testing was not completed this issue would not be

discovered until the configuration stage and the lead-time on replacement could cause serious delays to the overall delivery of the project.

We would recommend that a period of at least two weeks is allocated for testing of a new major site.

When initially devising a test plan, it's important to be mindful of the following:-

- Resources – What/Who/How many additional resources will be needed to complete a realistic series of tests and are they available at the appointed times?
- Equipment – What (if any) additional pieces of equipment are needed for testing?
- Lead-times – in the case of faulty/wrongly shipped equipment such as switches, power supplies, fiber modules etc how long will it take to receive replacement equipment and how does that fit with the go-live timetable?
- Problem resolution - have you allowed sufficient time to resolve any issues that might arise and is the expertise required to fix these issues available?

Specific tests

Assuming that initial equipment functionality tests are completed and verified and all equipment is installed and configured we would strongly recommend the following areas are rigorously tested (this list is by no means exhaustive):

- Internet links – where multiple links are installed, test both links individually.
Does failover between the individual links work correctly?
Does the bandwidth and overall performance of the link meet expectations from the ISP?
- Fiber links – check that all buildings linked by fiber can communicate between themselves and the core switches.
Does the overall performance of the links meet expectations?
- CAT6 cabling – check ethernet sockets, do they all work?
Can machines connected on the same physical LAN communicate with each other?
Does the overall performance meet expectations?
- COMPLIED • Switches – can the core switches communicate all Edge switches and vice versa?
In the case of stacked switches does failure of one switch affect it's stack partners?
In the case of multiple core switches, does failover work correctly?
- UPS – is the UPS correctly configured (i.e not in bypass mode)?
Can it maintain power to all equipment for the required time?
- COMPLIED • VLANs – are all VLANs configured correctly?
Can machines in different VLANs communicate with each other and essential central services such as DHCP servers, DNS servers etc?
Can machines in secured VLANs – i.e Guests – communicate with machines in non-secured VLANs, and if so why?
- Firewalls – is the Jotun VPN fully accessible?
Are all centrally delivered services such as Matrix, Steelmaster etc available and is performance acceptable?
In the case of clustered firewalls, does failover work correctly?
- COMPLIED • Wireless – are access points correctly placed to maximise coverage and performance?
Does roaming between access points work correctly?